American Computer Science League

2022-2023 ● Contest 4 : Shorts Problem Solutions ● Senior Division

1. Graph Theory

$$\begin{vmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{vmatrix}^4 = \begin{vmatrix} 3 & 6 & 2 & 2 \\ 5 & 6 & 5 & 5 \\ 5 & 2 & 6 & 6 \\ 3 & 2 & 3 & 3 \end{vmatrix}$$

Summing the entries in the first row gives 13 paths of length 4 from A.

B. 13

2. Graph Theory

The original directed graph had 9 cycles from A:
ABCA, ABDCA, ABEA, ABECA, ABEDCA, AEA, AEBCA, AEBDCA, and AECA. When directed edge DE is added, 3 new cycles are created: ABCDEA, ABDEA, and ABDECA.

C. 3

3. Digital Electronics

The Boolean expression represented by the digital circuit is:

$$\overline{(\overline{A} \oplus B) + \overline{BC}}$$

$$\begin{aligned} \overline{(\overline{A} \oplus B) + \overline{BC}} &= \overline{\overline{A} \oplus B}\,\overline{\overline{BC}} \\ &= (\overline{\overline{A}} \oplus B)BC \\ &= (\overline{A}B + A\overline{B})BC \\ &= \overline{A}BBC + A\overline{B}BC \\ &= \overline{A}BC \end{aligned}$$

D. $\overline{A}BC$

4. Digital Electronics

Let R be the symbol for the rectangular gate.

A	B	C	ARBRC	A(ARBRC)	(ARBRC)C	A(ARBRC)+(ARBRC)C
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	0	1	1
1	0	0	0	0	0	0
1	0	1	1	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0

Therefore only 3 ordered triples make the circuit TRUE.
(0, 1, 1), (1, 0, 1) and (1, 1, 0)

C. 3

5. Assembly Language

It finds the first Pell number that is a perfect square after 0 and 1.
A Pell number is a(n) = 2 * a(n-1) + a(n-2).
They are 0, 1, 2, 5, 12, 29, 70, 169, 408.

D. 169